

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data integration, Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 20%

QUESTIONS: 10

Total Marks:20

Annexure A

CLASS TEST

Criteria	Understanding of Task (2)	Procedure & Execution (2.5)	Observation & Result (2)	Viva Voce / Conceptual Response (2)	Time Management & Presentation (1.5)	Total(10)
Weightage	20%	25%	20%	20%	15%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						
<i>Q6</i>						
<i>Q7</i>						
<i>Q8</i>						
<i>Q9</i>						
<i>Q10</i>						

Code of Conduct:

I N.Vijetha (192311258) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

N. Vijetha

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Task	20%	Clearly understands the given task/experiment without assistance	Understands task with minor clarification	Partial understanding; requires guidance	Unable to understand the task
Procedure & Execution	25%	Performs procedure accurately and independently with proper technique	Performs with minor errors	Requires guidance; makes procedural mistakes	Unable to execute the procedure
Observation & Result	20%	Records accurate observations and obtains correct results	Minor errors in observation or result	Incomplete or partially correct results	Incorrect or no results
Viva Voce / Conceptual Response	20%	Answers all questions confidently with strong conceptual clarity	Answers most questions with minor gaps	Limited responses; needs prompting	Unable to answer questions
Time Management & Presentation	15%	Completes task on time with neat and clear presentation	Completes with minor delay or presentation issues	Delayed or poorly presented work	Unable to complete on time

CLASS TEST

QUESTIONS:

Total Marks: 10x2 =20

Q1. Define Data Mining and list any two basic data mining tasks. Explain the steps involved in KDD process and justify the role of data preprocessing in it. [L2, L4]

Q2. What is data cleaning? Mention two common data quality issues. Explain methods to handle missing values and noisy data with suitable examples. [L1, L2, L3]

Q3. Define data integration and list two challenges associated with it. Explain how redundancy and inconsistency are handled during data integration. [L2, L4]

Q4. What is covariance analysis? List the two covariance formalisms [L2, L3]

Q5. Suppose two stocks A and B have the following values in one week: (2, 5), (3, 8), (5, 10), (4, 11), (6, 14). Question: If the stocks are affected by the same industry trends, will their prices rise or fall together and independent ?

Q6. Explain the architecture of a typical data mining system with a neat diagram. Discuss the role of each component and analyze how they interact to support knowledge discovery. [L2, L4]

Q7. Find covariance for following data set $x = \{4,5,6,7,9\}$, $y = \{8,3,7,5,6\}$

Q8. Describe the major steps in data pre-processing. Given a dataset with missing values, outliers, and inconsistent formats, propose a systematic pre-processing pipeline and justify the sequence of steps. [L3, L4, L5]

Q9. A hospital is analyzing patient **age** data to identify broad health trends. Because the raw data contains minor variations and noise, the analyst needs to perform local smoothing using **equal-frequency** (equi-depth) binning. **Sample Data Set (Ages):** 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70 [L2, L4]

Q10. Explain data discretization and its importance in data mining. Compare equal-width and equal-frequency binning with suitable examples, and evaluate their impact on data analysis. [L3, L4, L5]

CO2 - ATL3 CLASS TEST

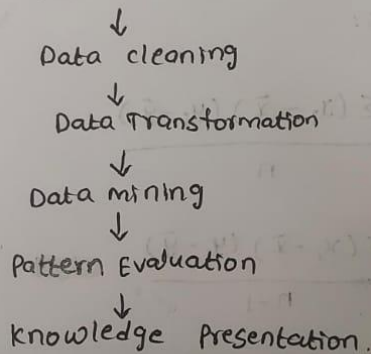
N. Vijetha
192311258
CSA1603
DWDM

1. **Data Mining:** It is a process of extracting useful patterns, relationships and knowledge from large amounts of data using statistical and database techniques.

⇒ Any two Data Mining Tasks:

1. **classification:** Assigns data into predefined categories.
2. **clustering:** Groups similar data objects together without predefined labels.

⇒ **steps in KDD:** Data selection



⇒ **Preprocessing:** Removes missing and noisy data.

2. **Data cleaning:**

It is a process of detecting and correcting inaccurate data.

⇒ Two common Data Quality Issues:

1. Missing values
2. Noisy Data

⇒ **Handling Missing values:**

- use mean, median, mode
- ignore the record
- predict using machine learning

⇒ **Handling Noisy data:**

- use Binning
- Regression
- clustering

3. Data Integration:

It combines data from multiple sources into one consistent dataset.

Challenges:

1. Data redundancy

2. Data inconsistency

Handling Redundancy:

- Remove duplicate records
- Correlation analysis
- Attribute matching

Handling Inconsistency:

- Standardize formats
- Resolve naming conflicts
- Apply validation rules

4. Covariance Analysis:

Covariance measures how two variables change together.

Two covariance formulas:

- Population:

$$\text{COV}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n}$$

- Sample:

$$\text{COV}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

5. Stock Covariance:

Given, Stock A = {2, 3, 5, 4, 6}

Stock B = {5, 8, 10, 11, 14}

$$\text{Mean}(A) = \frac{2+3+5+4+6}{5} \Rightarrow 4$$

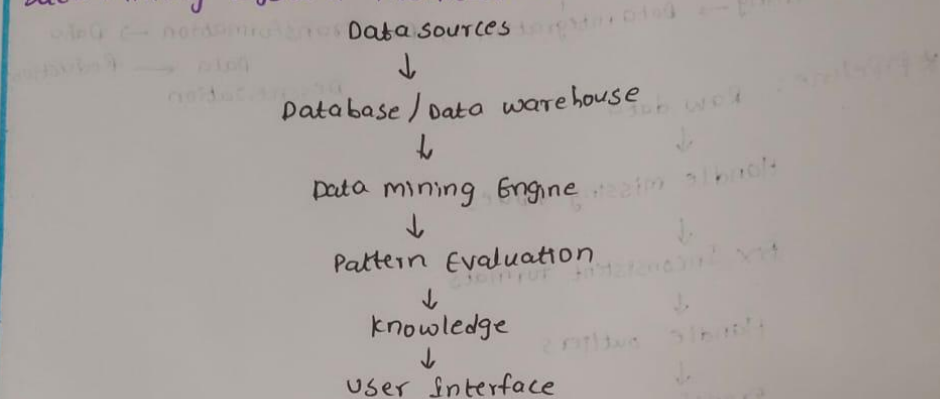
$$\text{Mean}(B) = \frac{5+8+10+11+14}{5} \Rightarrow 9.6$$

$\therefore \text{sum} = 20$

$$\text{Sample covariance} = \frac{20}{4} = 5 \text{ (positive)}$$

$\therefore$ Both stock prices rise and fall together, not independently.

6. Data Mining system Architecture:



Components:

- * Data sources: Databases, files.
- * Data base: stores integrated data
- * Data mining Engine: Performs classification, clustering.
- * Pattern Evaluation: selects useful patterns
- * Knowledge base: Stores domain knowledge
- * User Interface: Displays mining results.

Covariance calculation:

Given, $x = \{4, 5, 6, 7, 9\}$

$y = \{8, 3, 7, 5, 6\}$

Mean of x :

6.2

Mean of y :

5.8

Sum of deviation products: -1.8

Indicates an inverse relationship.

Population covariance:

$$\text{Cov}(x, y) = \frac{-1.8}{5} = -0.36$$

Sample covariance:

$$\text{Cov}(x, y) = \frac{-1.8}{4} = -0.45$$

8. Data Pre-processing major steps:

Data cleaning → Data Integration → Data Transformation → Data Reduction
Data ← Discretization

* Pipeline: Raw data

↓
Handle missing values
↓
Fix inconsistent formats
↓
Handle outliers
↓
Transform data

Result: Produces clean data for data mining.

9. Equal-frequency Binning:

* Dataset: contains 27 age values

* Bin 1: 13, 15, 16, 16, 19, 20, 20, 21, 22

Mean = 18

* Bin 2: 22, 25, 25, 25, 25, 30, 33, 33, 35

Mean = 28.11

* Bin 3: 35, 35, 35, 36, 40, 45, 46, 52, 70

Mean = 43.78

∴ Results: {18, 18, 18, 18, 18, 18, 18, 18, 18, 28.11, 28.11, 28.11, 28.11, 28.11, 28.11, 28.11, 28.11, 28.11, 43.78, 43.78, 43.78, 43.78, 43.78, 43.78, 43.78, 43.78, 43.78, 43.78}

10. Data Discretization: It converts continuous numerical values into interval bins.

Importance: • Reduces noise • simplifies data analysis

Equal width: Divides the data into bins of equal size

Equal-frequency: Each bin contains the same number of data values.